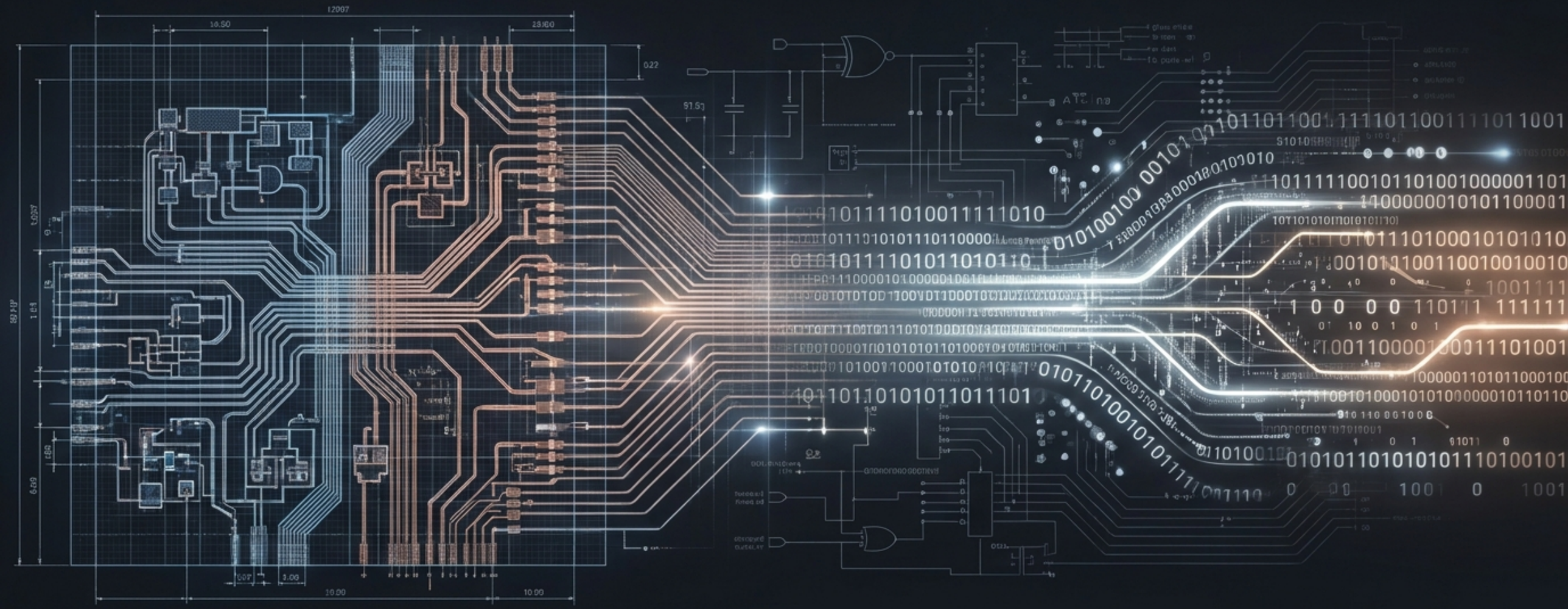
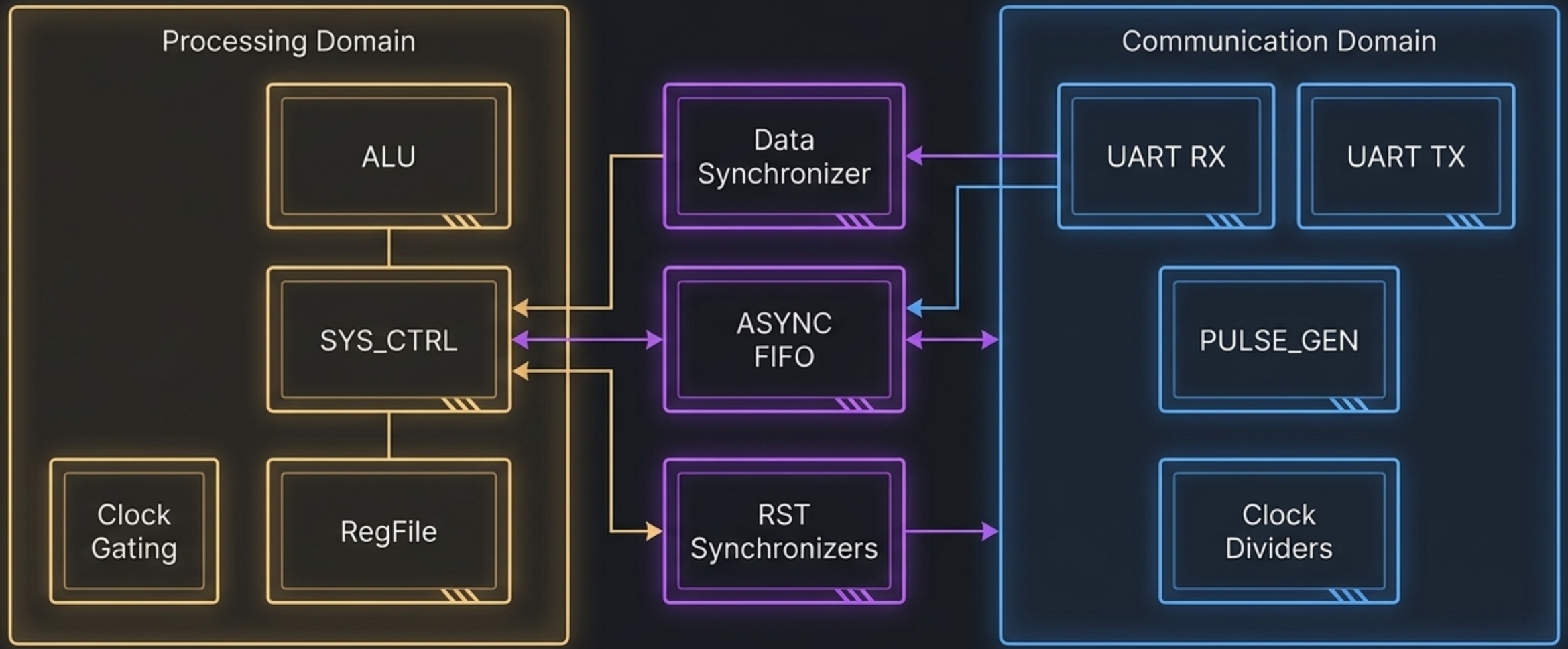
UART Controlled Digital System

Architecture, Implementation, and Physical Verification

Target Node: TSMC 130nm
(scmetro_tsmc_cl013g_rvt)



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Processing Domain

Clock Source: REF_CLK (50 MHz)

Timing Nature: Fully Synchronous

Workload: Fast execution (System Control, ALU operations, Register File read/writes)

Power Strategy: Power-gated (Clock Gating activates only during ALU_FUN/EXECUTION)

Communication Domain

Clock Source: UART_CLK (3.6864 MHz)

Timing Nature: Asynchronous / Serial I/O

Workload: Slow data ingress/egress (UART TX/RX, Baud rate generation)

Power Strategy: Continuous operation via divided clocks

The Chasm

Requires robust Data, Reset, and Async FIFO synchronization bridges to guarantee data integrity across boundaries.

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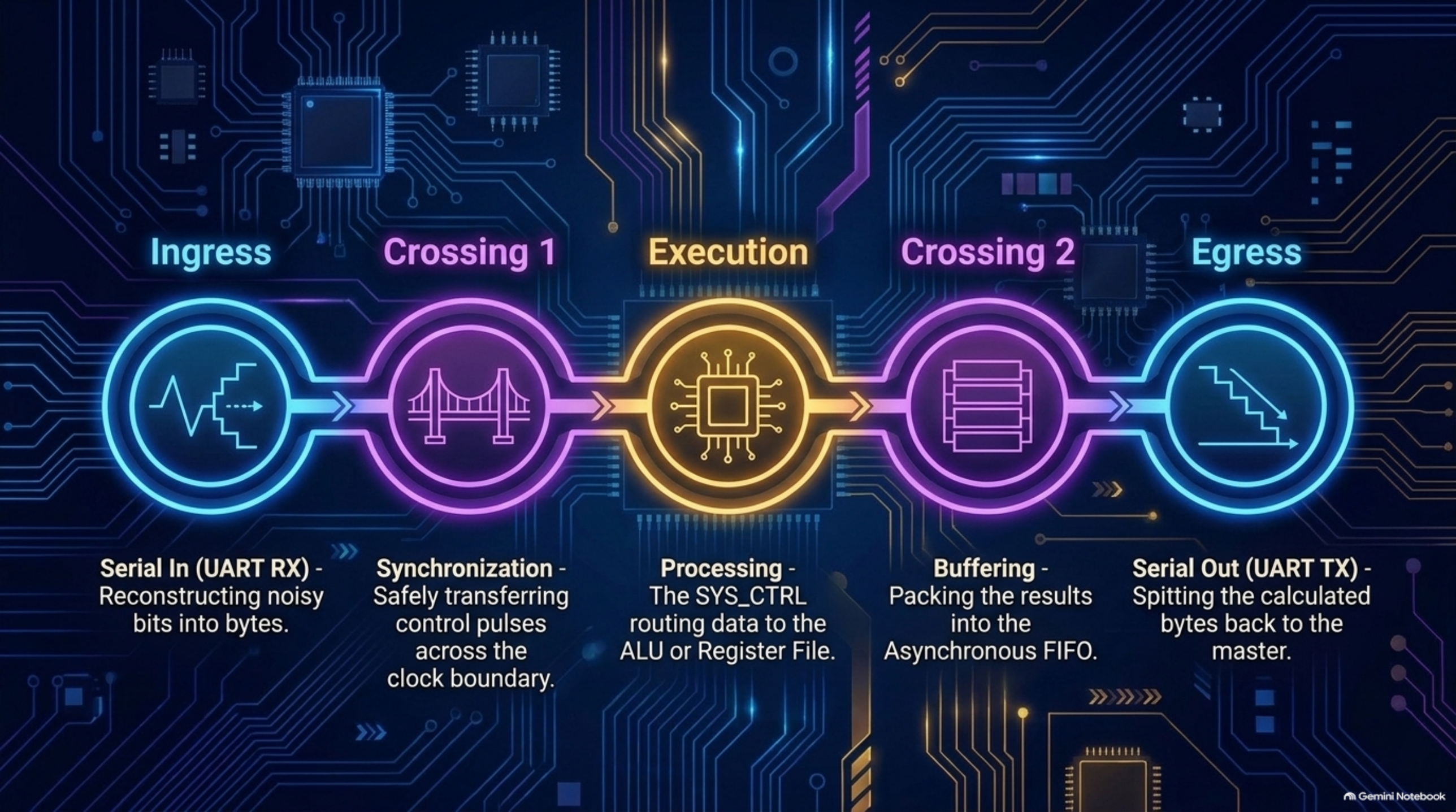
CLOCK DISTRIBUTION NETWORK



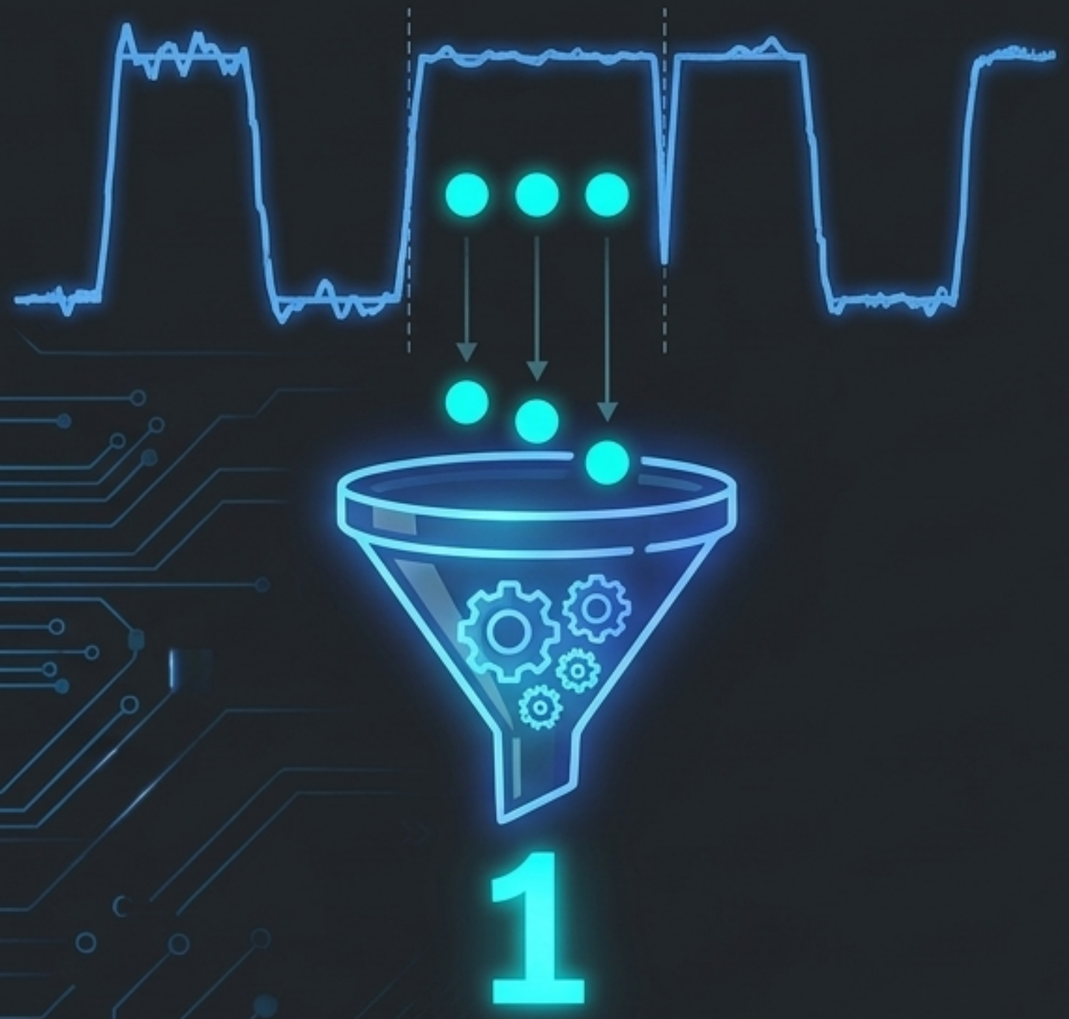


Clock Gating Cell
(captures clk_en while REF_CLK
is low to avoid truncation)

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NOISE FILTERING & BIT RECOVERY: OVERSAMPLING & MAJORITY VOTE

Oversampling Architecture

Supports prescale values of 8, 16, and 32.

Majority Vote Resolution

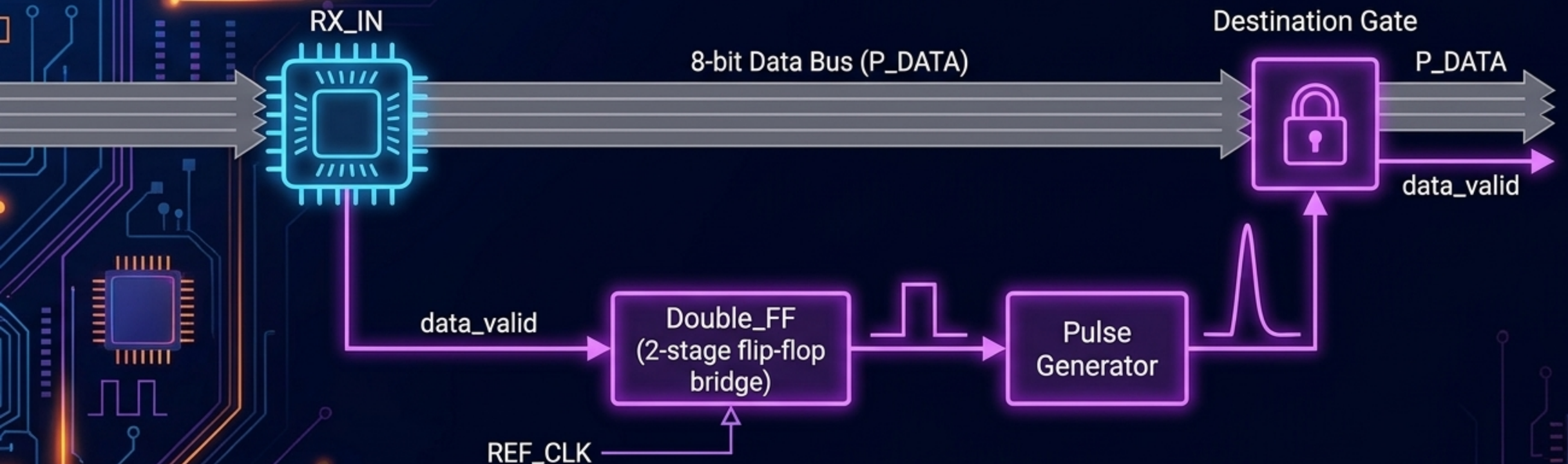
The sampler takes 3 observations (at P/2-1, P/2, P/2+1). The hardware chooses the value present in at least two of the three stored samples, neutralizing transient line noise.

FSM Data Validation

Dedicated checker blocks perform parallel verification on the fly: Start glitch detection, Parity accumulation (even/odd configuration), and Stop bit validation.

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Multi-Cycle Path



Control-Level Synchronization

Rather than placing synchronizer chains on every data bit (which risks skew), the system synchronizes the enable signal.

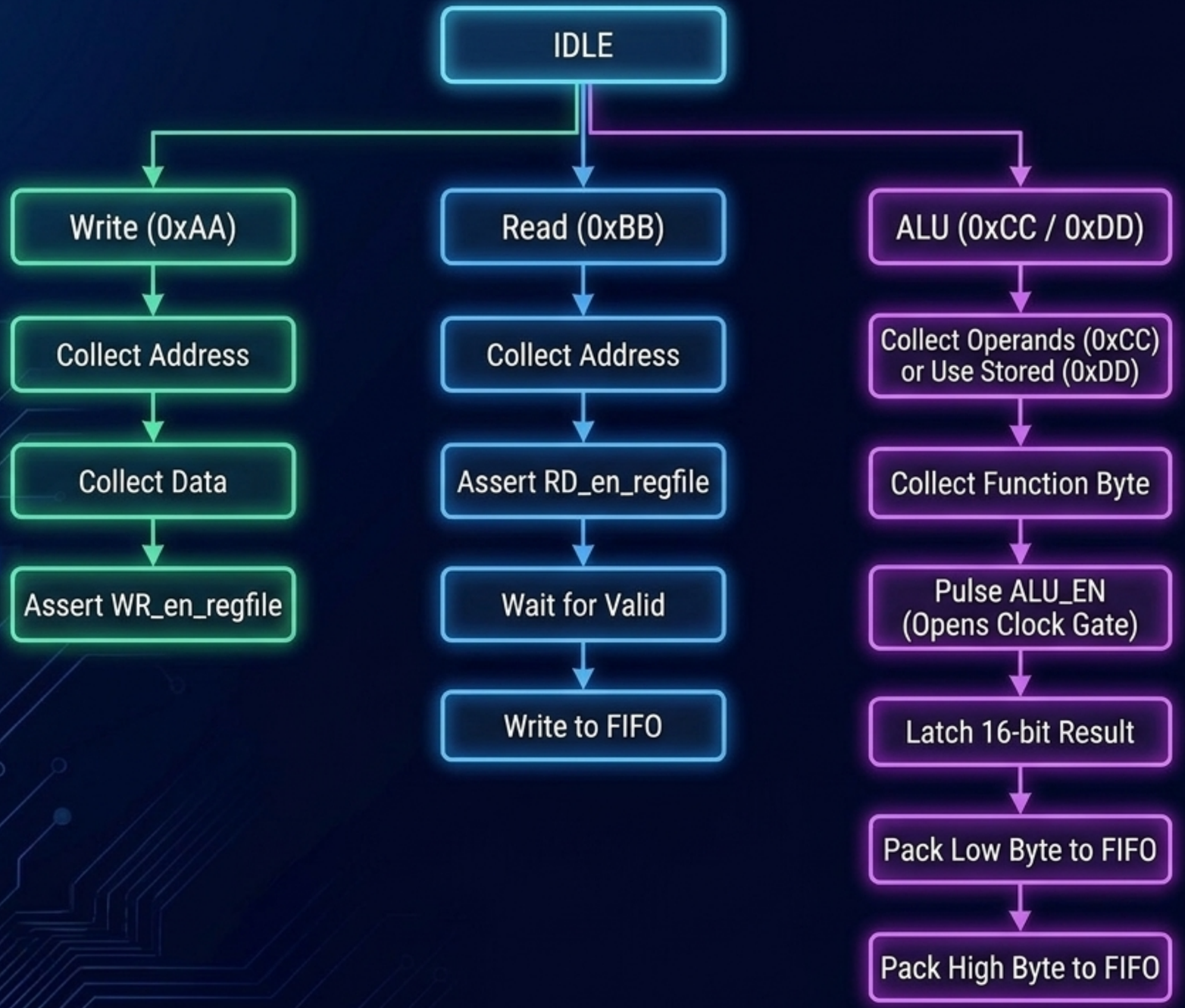
The Bridge

A 2-stage flip-flop bridge running on REF_CLK mitigates metastability.

Pulse Generation

A dedicated edge detector converts the synchronized enable into a single destination event, safely latching the stable unsynchronized bus into the processing domain.

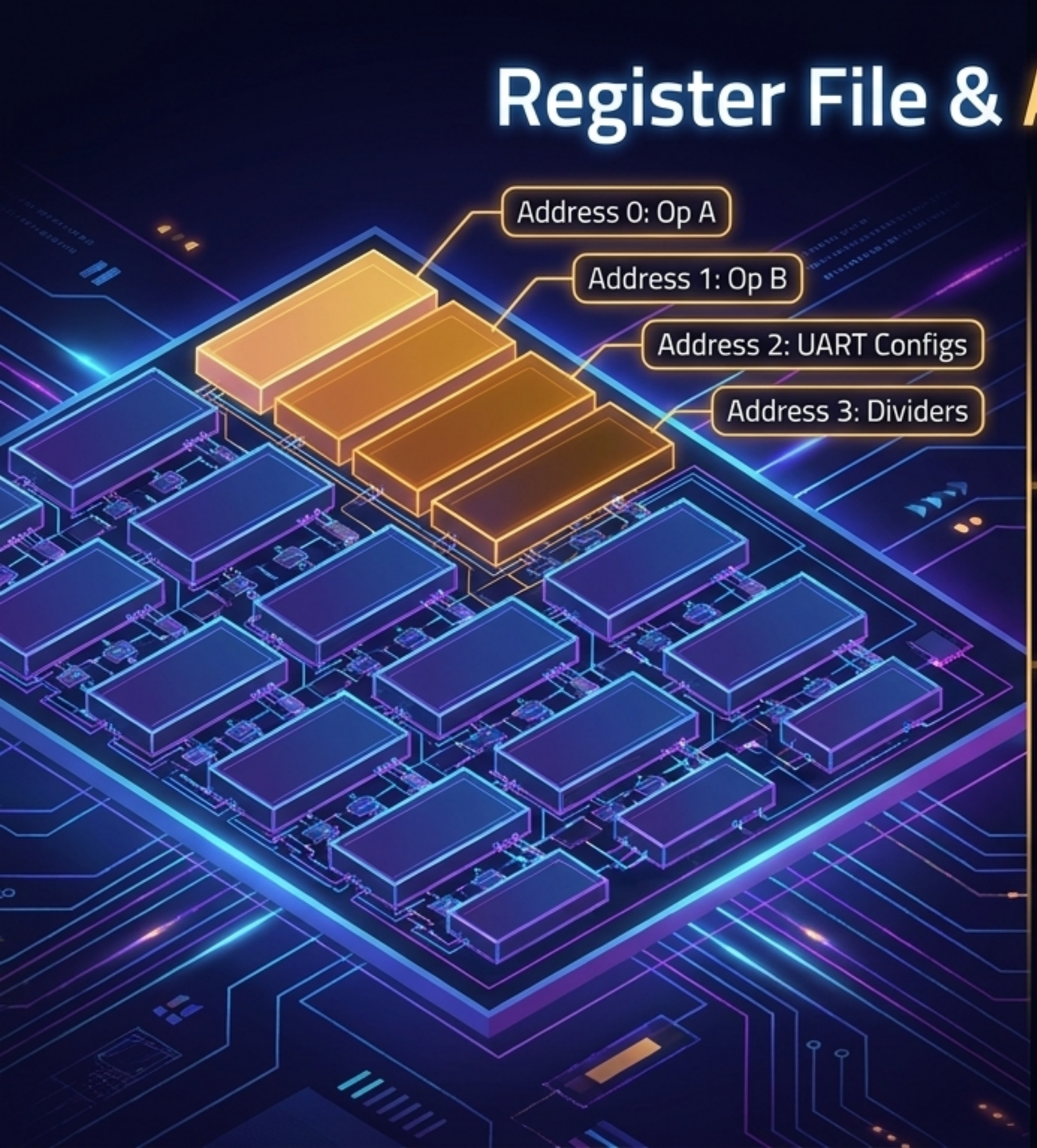
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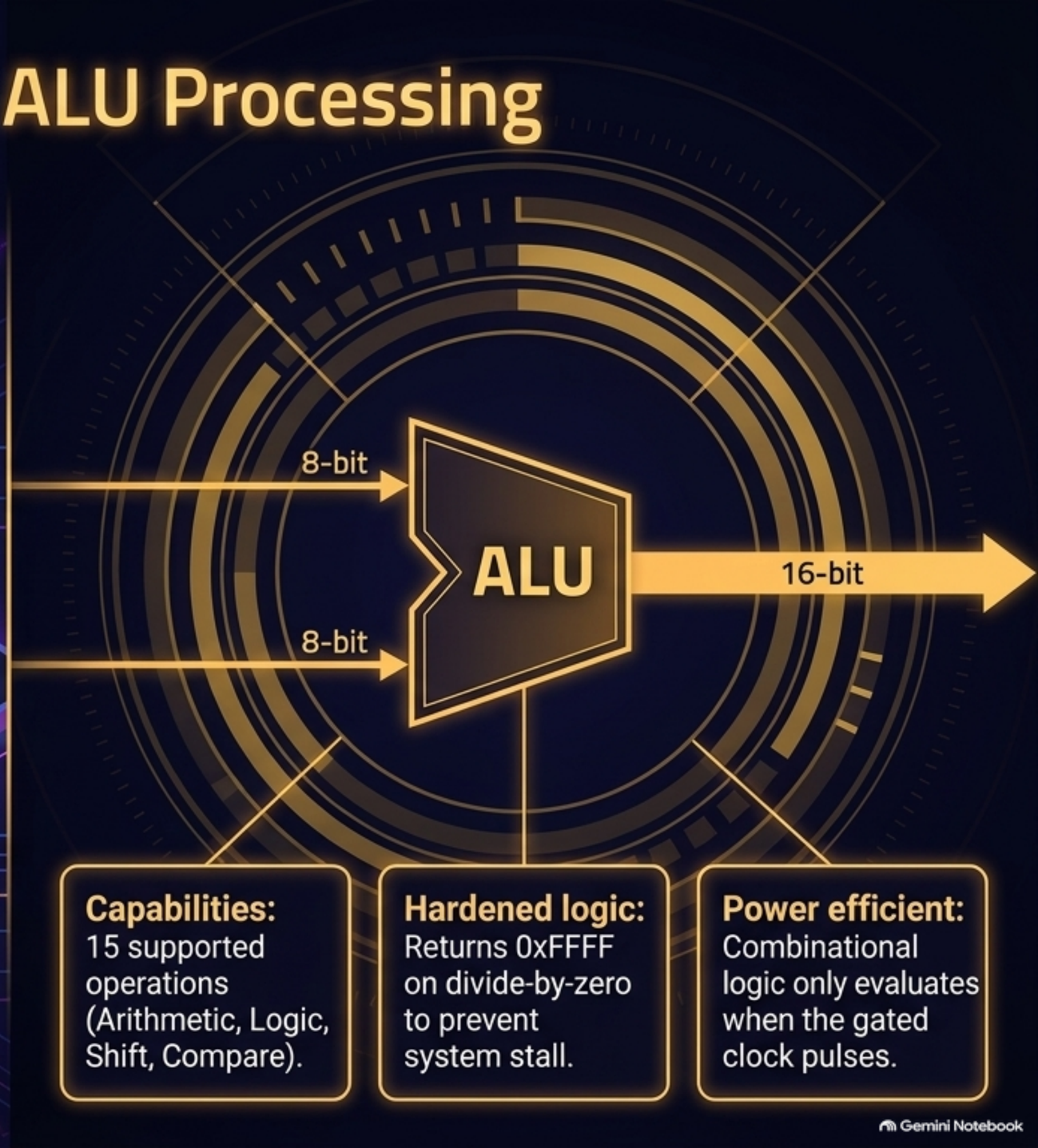


Backpressure Aware:
The FSM holds in WRITE_IN_FIFO if the Async Async FIFO asserts a full flag, protecting response storage.

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Register File & ALU Processing



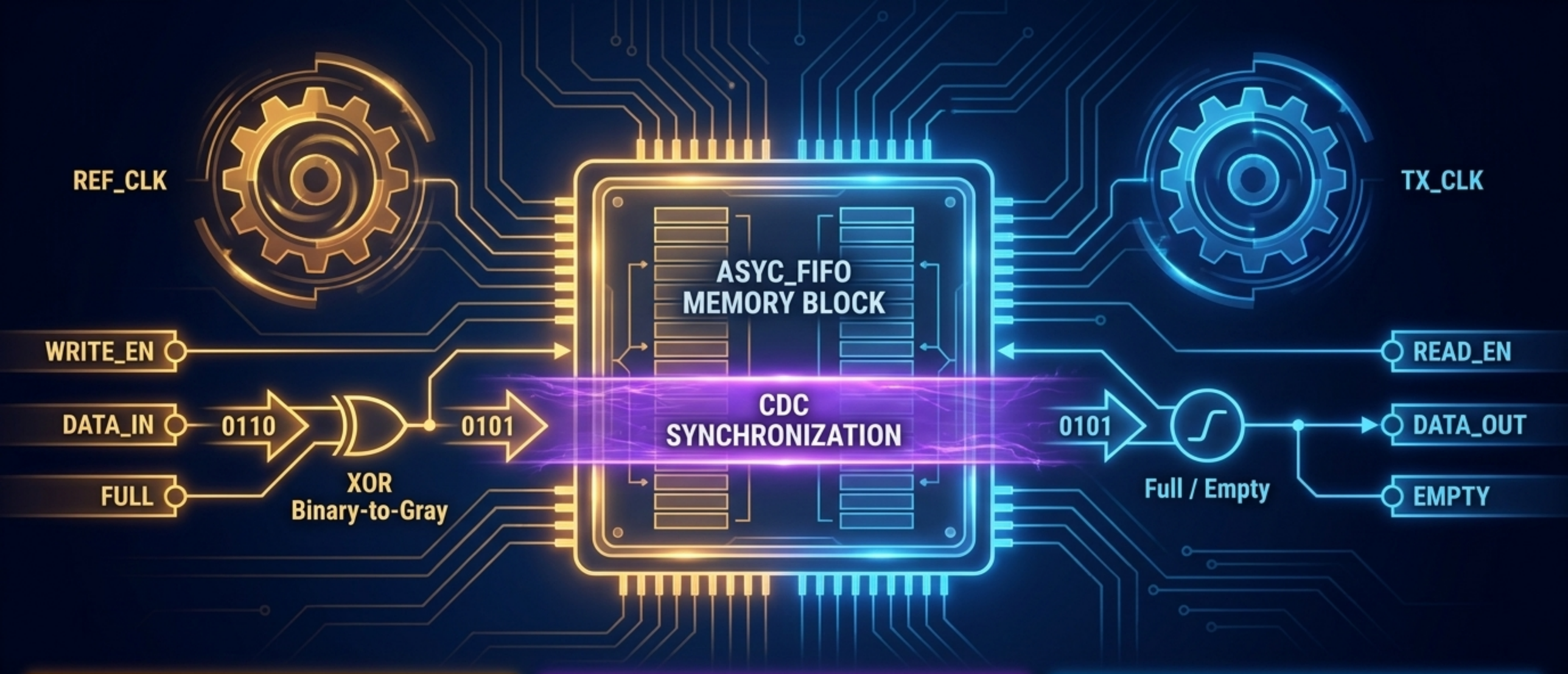


Capabilities: 15 supported operations (Arithmetic, Logic, Shift, Compare).

Hardened logic: Returns 0xFFFF on divide-by-zero to prevent system stall.

Power efficient: Combinational logic only evaluates when the gated clock pulses.

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Dual-Port RAM Architecture

Written strictly on REF_CLK, Read strictly on TX_CLK.

Binary-to-Gray Conversion

Binary pointers are converted to Gray code before synchronization. Because Gray code only changes one bit at a time, it physically prevents multi-bit transition errors during CDC.

First-Word-Visible

The read interface exposes the head byte continuously, providing immediate backpressure to the controller when full.

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UART Transmission Sequence Diagram



Dequeue Initiation

Transmission triggers automatically when the FIFO ~empty flag drops.

Serialization

Data is shifted out Least-Significant-Bit first.

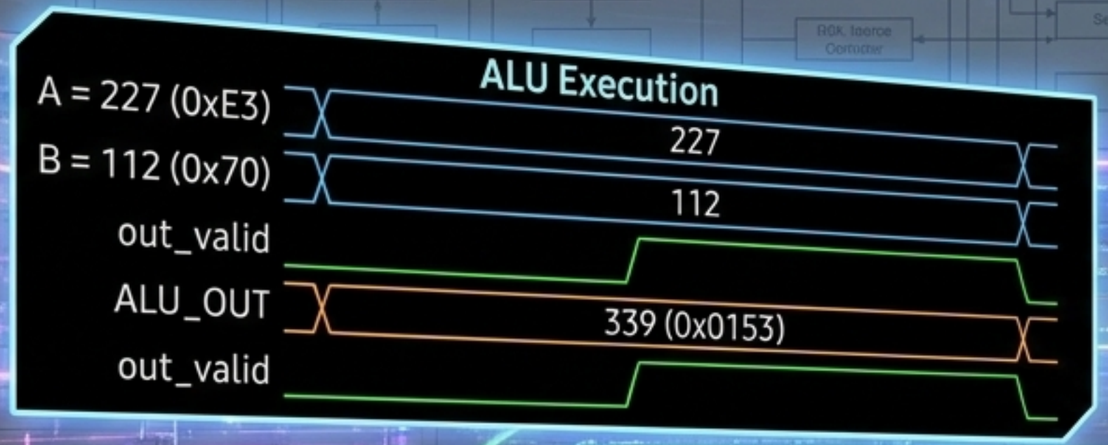
On-the-Fly Parity

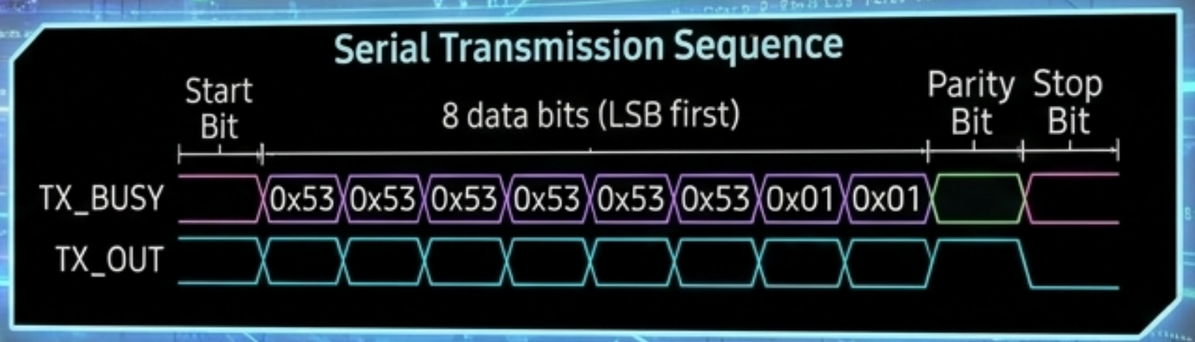
The transmission sequence dynamically calculates and appends Even or Odd parity based on the live REG2 configuration.

FIFO Pop

The read pointer advances at the start of transmission, freeing space in the FIFO immediately.

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Comprehensive ALU Coverage

70 randomized ALU operations successfully verified across 15 operation types.

Waveform Proof

Graphic perfectly highlights the execution: 227 + 112 = 339 (0x0153), with the transmission showing the low byte (0x53) followed by the high byte (0x01).

Robust Error Injection

Assertions proved that the system correctly catches and flags simulated Parity and Stop bit errors without corrupting the FSM.

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Synthesis Specs

Target Library: scmetro_tsmc_cl013g_rvt (130nm Node).

Pre-DFT Area footprint: 21,700 library units.

Design For Testability (DFT)

Full scan methodology utilizing multiplexed flip-flops.

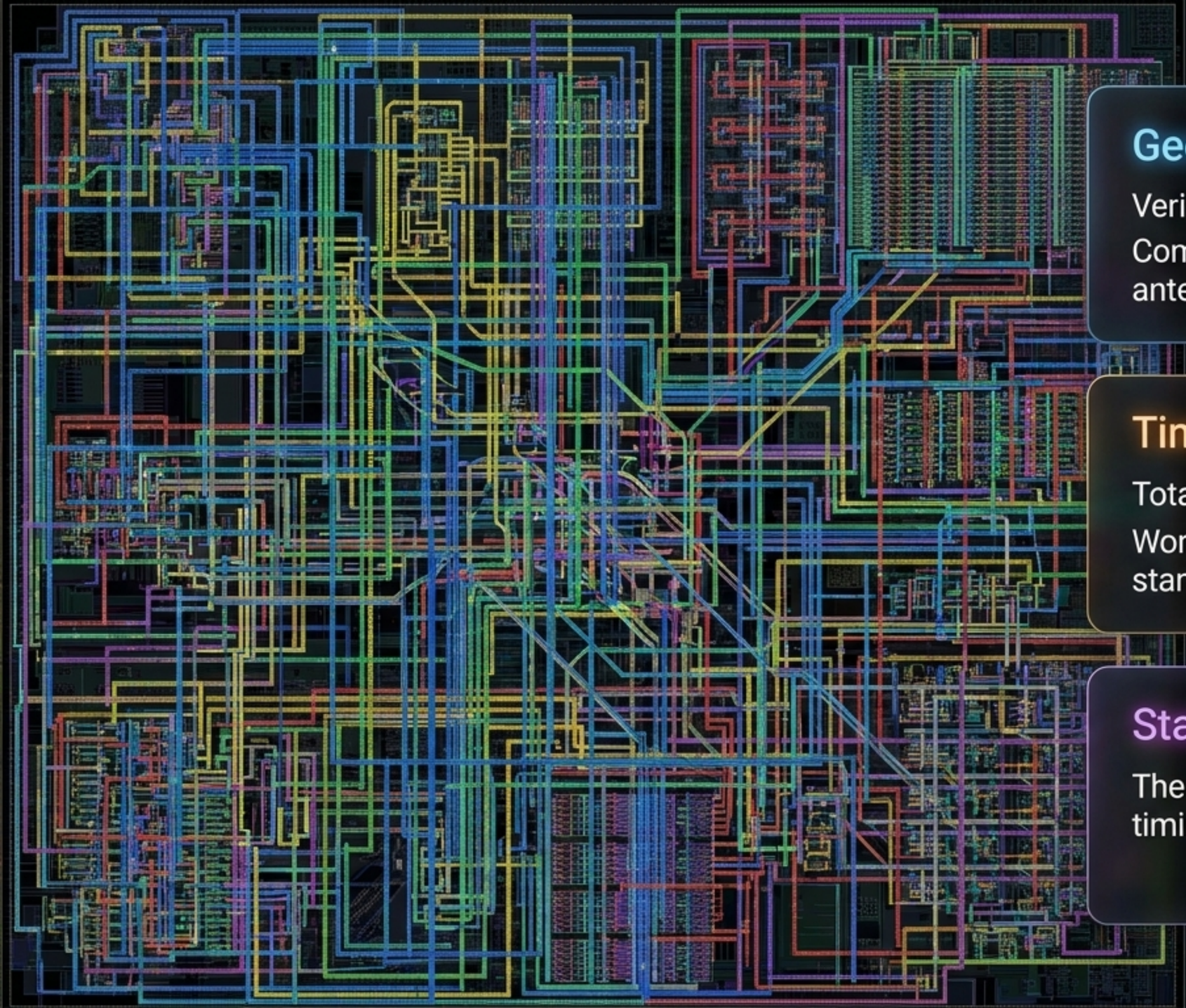
356 scan cells successfully organized into 4 preview chains.

Equivalency Proof

Synopsys Formality achieved 100% verification—all 360 compare points perfectly matched between the behavioral RTL and the synthesized netlist.

```
+---------------------------------------+
| AREA REPORT FOR DESIGN : top_system   |
| (Pre-DFT)                             |
+---------------------------------------+
| Combinational area     : 18648.000000 |
| Noncombinational area  : 3156.0000000 |
| Buf/Inv area           : 2106.0000000 |
| Net Interconnect area  : 15602.000000 |
| Total cell area        : 21804.000000 |
| Total area             : 37406.000000 |
+---------------------------------------+
| Cell Count             : 4134         |
| Leaf Cell Count        : 4134         |
+---------------------------------------+
Design is not fully mapped.
```

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Geometry & Connectivity

Verified 0 Design Rule Violations (DRVs).

Complete routing without shorts or antenna violations.

Timing Closure Achieved

Total setup cost is zero.

Worst Negative Slack (WNS) for setup stands at a healthy +5.576 ns.

Status

The digital logic is physically viable, timing clean, and ready for tape-out.

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Extreme Disparity Handled

Successfully managed asynchronous communication bridging a 3.6MHz serial domain and a 50MHz processing core.

Fault-Tolerant Boundaries

Implemented mathematically robust Binary-to-Gray Async FIFOs and Multi-Cycle Path synchronizers.

Silicon Proven

Fully verified end-to-end—from RTL behavioral testbenches, through DFT scan insertion, to a timing-closed, zero-violation Physical Layout.

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